

EventBrain AI: Intelligent Event-Saver

QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

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Project Name: EventBrain AI - Intelligent Event-Saver

Version: 2.0

1. Project Overview

- **Problem Statement:**

Corporate event planning in Malaysia remains highly manual, fragmented, and reactive. Medium to large companies often organize 30–50+ internal and external events annually, including team buildings, festive celebrations, workshops, product launches, and award ceremonies.

Event coordinators currently rely on spreadsheets, WhatsApp chats, emails, and manual vendor follow-ups to manage planning workflows. This creates major operational inefficiencies such as:

- Vendor double bookings
- Slow vendor negotiations
- Budget overruns
- Poor coordination between departments
- Delayed decision-making
- Lack of real-time risk detection (weather issues, public holidays, vendor cancellations)

Many event requests also arrive in unstructured formats such as:

"Plan a team building event for 80 employees next Friday with RM4,000 budget."

Traditional workflows struggle to interpret these requests, assess feasibility, and respond quickly to unexpected disruptions. As a result, companies lose 20–30 working

hours per event, experience employee burnout, and miss opportunities to maximize event ROI.

- **Target Domain:**

AI System & Agentic Workflow Automation for Corporate Event Operations

The platform focuses on automating the end-to-end workflow of corporate event planning, vendor coordination, risk management, and crisis response.

- **Proposed Solution Summary:** EventBrain AI is a Gemini-powered multi-agent event orchestration platform that automates corporate event planning from initial request intake to crisis recovery.

Users simply describe their event requirements in natural language, and the system autonomously coordinates multiple AI agents to:

- Extract event requirements from unstructured requests
- Identify unrealistic budgets/timelines
- Recommend suitable vendors
- Draft vendor communication emails
- Detect scheduling conflicts and external risks
- Generate execution plans based on cost/risk tradeoffs
- Respond instantly to vendor cancellations or sudden disruptions

Instead of spending hours coordinating vendors, budgets, and contingency planning manually, event coordinators shift from being overwhelmed planners to high-level decision makers who simply review AI-generated recommendations and approve actions.

EventBrain AI transforms event planning from a fragmented manual process into an intelligent, autonomous workflow system that saves time, reduces operational risk, and improves event execution efficiency.

2. Background & Business Objective

- **Background of the Problem:** Corporate event planning is still highly manual and fragmented, relying on spreadsheets, emails, and messaging tools to coordinate vendors, budgets, schedules, and logistics. For companies running 30–50+ events yearly, coordinators must repeatedly handle unstructured requests like:

"Plan a team building for 80 people next Friday with RM4,000 budget."

This requires manual interpretation, vendor sourcing, risk checking, and coordination across multiple tools. When disruptions occur (vendor cancellation, weather, scheduling conflicts), the process becomes reactive and error-prone.

As a result, organizations face:

- High planning time per event (20–30 hours)
 - Budget inefficiencies
 - Coordination delays
 - Operational burnout
 - Reduced event quality
- **Importance of Solving This Issue:** Automating event planning significantly reduces manual coordination effort and improves decision-making speed.

EventBrain AI enables:

- Faster event planning (minutes instead of hours/days)
- Early detection of budget and scheduling risks
- Reduced human coordination workload
- Faster response to disruptions
- Better resource and vendor selection

This allows companies to execute more events efficiently while improving overall productivity and ROI.

- **Strategic Fit / Impact:** EventBrain AI aligns with the shift toward AI-driven workflow automation and agentic systems.

It supports organizations by:

- Automating repetitive operational workflows
- Improving decision-making through AI reasoning
- Enhancing efficiency in HR and event operations
- Demonstrating real-world application of multi-agent AI systems

It positions itself as an AI event operations layer that transforms traditional manual planning into an intelligent, automated workflow.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-powered event planning assistant that automates corporate event workflows from planning to crisis management. The goal is to reduce manual coordination work and enable faster, smarter event execution.

3.2. Intended Users (Target Audience):

- HR Departments
- Event Coordinators

4. System Functionalities

4.1. Description: EventBrain AI operates as an AI-powered event orchestration platform that transforms unstructured event requests into actionable execution plans through multiple specialized AI agents. Users can describe their event needs in natural language, and the system automatically handles requirement extraction, vendor matching, risk detection, decision support, and crisis recovery. The platform reduces manual coordination by automating repetitive planning workflows while keeping humans involved for final approvals.

4.2. Key Functionalities:

- **Smart Event Intake:** Converts natural language event requests into structured event requirements such as budget, location, timeline, and participant count.
- **Budget Feasibility Analysis:** Detects unrealistic budgets, timelines, or requirements and recommends better alternatives.
- **Vendor Matching & Ranking:** Identifies suitable vendors based on price, availability, and performance ratings.
- **Automated Vendor Communication:** Generates ready-to-send vendor inquiry or replacement emails automatically.
- **Risk Detection:** Identifies scheduling conflicts, public holidays, weather risks, and event-related disruptions.
- **Decision Planning:** Generates multiple execution strategies based on cost, speed, and risk trade-offs.
- **Crisis Response Automation:** Automatically reacts to vendor cancellations or sudden disruptions by generating backup plans in real time.

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

EventBrain AI uses Gemini API as its core language model.

It was selected because it offers:

- Strong natural language understanding for messy event requests
- Good reasoning capabilities for budget/risk evaluation
- Fast response time for real-time workflow execution
- OpenAI-compatible API integration for easier deployment
- Cost efficiency for multi-agent workflows compared to larger premium models

These capabilities make Gemini suitable for handling event planning tasks that require both structured extraction and decision-making.

4.3.2. Prompting Strategy:

The system uses multi-step agentic prompting.

Instead of relying on a single prompt, each AI agent is assigned a specialized task:

- Intake Agent → extracts event requirements
- Vendor Agent → finds vendors and drafts emails
- Risk Agent → checks conflicts and risks
- Decision Agent → generates planning strategies
- Crisis Agent → handles disruptions

This approach improves accuracy, reduces prompt overload, and makes the workflow easier to scale.

We also use few-shot prompting for vendor email generation to maintain consistent communication formatting.

4.3.3. Context & Input Handling:

The system accepts natural language inputs such as event requests, emails, and vendor Messages.

Example:

"Need a team building event for 80 people next Friday with RM4,000 budget."

For oversized inputs:

- Maximum accepted input: based on GLM token limit
- Long emails/documents are chunked into smaller sections
- Important details are extracted and summarized before processing
- Extremely irrelevant or unsupported files are rejected with user feedback

This ensures stable performance without overwhelming the model.

4.3.4. Fallback & Failure Behavior:

To reduce hallucinations and unusable outputs:

- Vendor recommendations are validated against database records
- Calendar conflicts are verified through external APIs
- Risk checks use rule-based verification layers
- If outputs are unclear, the system retries prompts once

If failure persists:

- The system flags the issue
- Escalates the task to human review
- Displays fallback recommendations instead of fully automated action

This ensures reliability while maintaining human oversight for critical decisions.

5. User Stories & Use Cases

- **User Stories:**

- As an HR manager, I want to describe event requirements in natural language so I can plan events faster without manually coordinating everything.
- As an event coordinator, I want AI to recommend suitable vendors so I can reduce time spent comparing options.
- As an operations manager, I want the system to detect scheduling and budget risks early so I can avoid costly mistakes.
- As a company admin, I want AI-generated vendor emails so I can speed up communication.
- As an event planner, I want instant backup plans when vendors cancel so my event can continue without major delays.

- **Use Cases (Main Interactions):**

Event Planning Flow:

User enters event request in chat:

"Need a team building event for 80 people next Friday, RM4,000 budget in Klang Valley."

System workflow:

- Agent 1 extracts event details
- Flags budget issues
- Creates structured event brief

Vendor Recommendation Flow:

After intake:

- Agent 2a filters vendors from database
- Ranks top vendors using GLM
- Displays recommended vendors

Risk Detection Flow:

Agent 2b:

- Checks holidays
- Detects peak season conflicts
- Evaluates weather and timeline risks
- Returns mitigation suggestions

Vendor Communication Flow:

Agent 3:

- Generates ready-to-send inquiry emails for selected vendors

Decision Planning Flow:

Agent 4:

- Generates conservative, balanced, and aggressive execution plans for user approval.

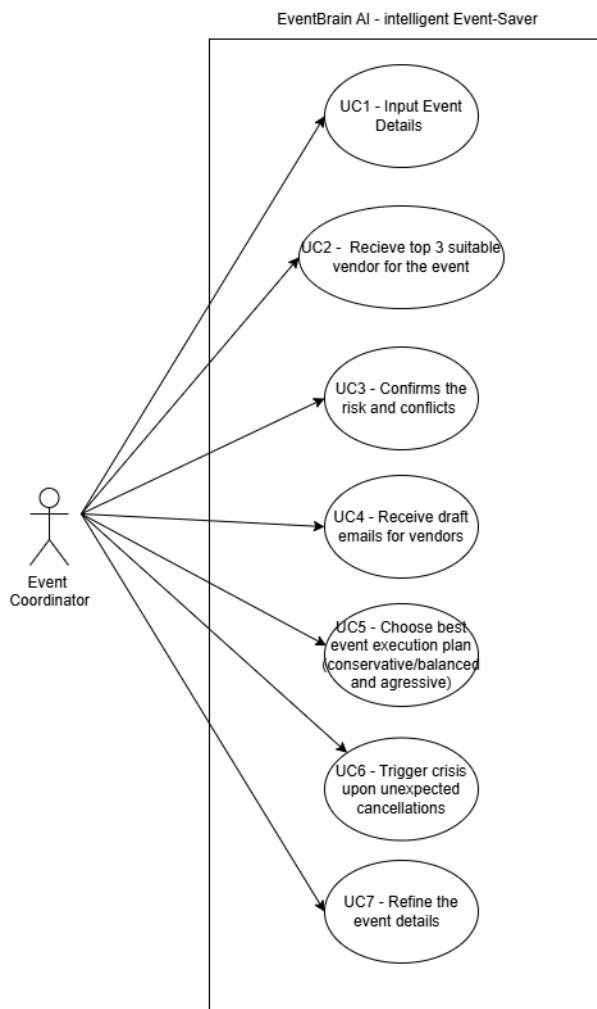
Crisis Recovery Flow:

User clicks crisis trigger:

"Vendor cancelled 2 days before event."

Agent 5:

- Removes cancelled vendor
- Finds replacements
- Drafts urgent emails
- Generates revised event plan

• Use Case Diagram:

6. Features Included (Scope Definition)

- Natural language event request input (Malay, English, Manglish supported)
- AI event requirement extraction (budget, pax, location, date, event type)
- Budget feasibility analysis and budget-per-pax warnings
- Vendor database filtering and intelligent vendor ranking
- Public holiday and peak season conflict detection
- Weather and lead-time risk assessment
- AI-generated vendor inquiry emails
- Multiple execution plan recommendations (conservative, balanced, aggressive)
- Real-time agent workflow tracking dashboard
- Crisis simulation for vendor cancellation
- Automatic replacement vendor recommendations
- Emergency email drafting for replacement vendors
- Human approval before sending emails/actions

7. Features Not Included (Scope Control)

- Real payment processing
- Full live vendor marketplace integration
- Real contract negotiation automation
- SMS/WhatsApp vendor integrations
- Multi-company multi-tenant deployment
- Real-time booking confirmation with external vendors
- Full enterprise ERP integration
- Automatic event execution without human approval
- Mobile application version
- Advanced analytics/reporting dashboard
- Multi-user collaboration for the same event session (MVP limitation)

8. Assumptions & Constraints

- **LLM Cost Constraint:**

Each user session typically triggers multiple Gemini calls across agents (Intake, Vendor, Risk, Email, Decision, Crisis simulation).

Estimated usage per session:

- ~5–8 LLM calls per full workflow
- ~2,000–4,000 tokens per session (average event request + agent outputs)

To control inference cost at scale, the system is designed with:

- Pre-filtering using Python before LLM calls (reduces unnecessary reasoning load)
- Parallel execution of Agent 2a & 2b (reduces redundant sequential calls)
- Database-first vendor filtering before Gemini ranking (LLM only ranks shortlisted vendors)
- Caching repeated vendor queries and holiday checks

These design decisions ensure Gemini is only used for reasoning tasks, not raw filtering or retrieval.

- **Technical Constraints:**

- Vendor database is mock/structured (Supabase), not a real live marketplace
- Email integration is limited to simulation or sandbox Gmail API
- Calendar conflict detection relies on predefined holiday dataset (Malaysia-focused)
- Weather data is assumed/static for hackathon demo scope
- System is optimized for single-session event planning (not multi-tenant enterprise scale)

- **Performance Constraints:**

- Each full workflow depends on multiple sequential + parallel agent calls
- Response time increases with number of agents triggered (especially Decision + Crisis agents)
- Crisis mode re-runs multiple agents, increasing latency

- Token-heavy outputs (execution plans + emails) are limited to 3 variants per response
- Real-time UI updates depend on WebSocket performance and backend load
- **User Input:**
 - System requires explicit user prompt to initiate event planning
 - No autonomous event creation without user input
 - Human approval is required before sending vendor emails or finalizing execution plans
 - System is not fully production-automated and acts as a decision-support assistant

9. Risks & Questions Throughout Development

- **Vendor Data Reliability:** How do we ensure vendor data (pricing, availability, ratings) remains accurate if the database is mocked or outdated?
- **Hallucination in Vendor Recommendations:** How do we prevent Gemini from recommending vendors inconsistently or inventing unrealistic pricing/suitability?
- **Agent Dependency Failures:** What happens if one agent (e.g. Risk Agent or Vendor Agent) fails or returns incomplete data? Does the system degrade gracefully?
- **Crisis Recovery Accuracy:** How reliable is the crisis response agent in selecting appropriate replacement vendors under time pressure constraints?
- **Multi-Agent Latency:** Does parallelizing agents (2a and 2b) sufficiently reduce latency, or will chained dependencies still slow down the system in real scenarios?
- **Human-in-the-Loop Dependency:** Is requiring human approval before execution a limitation to full automation, or a necessary safeguard for enterprise use?